

Seminar 5.

Def Domain of the function - the set of inputs accepted by the function. In order to say it is "what x can be in function"

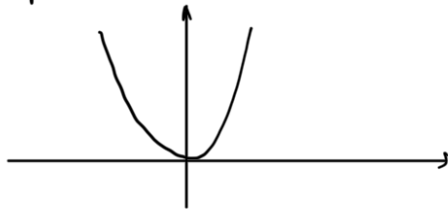
for example: if we have $f(x) = \sqrt{x} \Rightarrow$

$$\text{domain : } D(f) : x \in [0; +\infty)$$

Range - the set of all possible outputs

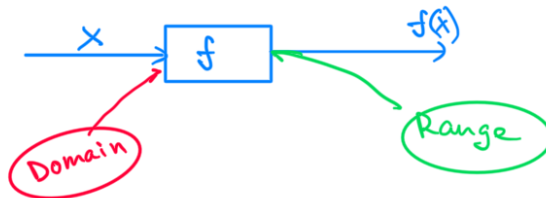
$$f(x) = \sqrt{x} \Rightarrow R(f) : f(x) \in [0; +\infty)$$

Example: $f(x) = x^2 :$



$$D(f) = (-\infty; +\infty)$$

$$R(f) = [0; +\infty)$$



Problem 1.

$$a) f(x) = \frac{1}{1+x}$$

$$1+x \neq 0$$

$$x \neq -1$$

$$D(f) = (-\infty; -1) \cup (-1; +\infty)$$

$$y = \frac{1}{1+x}$$

$$1+x = \frac{1}{y}$$

$$x = \frac{1}{y} - 1$$

$$R(f) = (-\infty; 0) \cup (0; +\infty)$$

$$b) f(x) = \begin{cases} |1-x|, & x < 2 \\ \sqrt{x+2}, & x \geq 2 \end{cases}$$

$\swarrow h(x)$ $\swarrow g(x)$

$$D(f) : \begin{cases} \bullet |1-x| \rightarrow \text{make sense } \forall x \\ \bullet \sqrt{x+2}, \quad g(x) > 0, \quad \text{as by the cond.} \\ \quad x > 2 \Rightarrow \text{no restriction.} \end{cases}$$

$$D(f) = (-\infty; +\infty)$$

$$R(f) : \int |h(x)| = \text{graph of } |1-x| \rightarrow y \geq 0$$

$$\sqrt{g(x)} \Rightarrow y \geq 0$$

$$\Rightarrow R(f) = [0; +\infty).$$

Problem 3.

Def (Cauchy ϵ - δ)

$$\lim_{x \rightarrow x_0} f(x) = L \Leftrightarrow \forall \epsilon > 0 \quad \exists \delta > 0 \quad \forall x: \underline{0 < |x - x_0| < \delta} \Rightarrow$$

$$\Rightarrow |f(x) - L| < \epsilon$$

how to prove: • arbitrary $\epsilon > 0$

• must present $\delta > 0$ (it may depend on ϵ)

• check $0 < |x - x_0| < \delta \Rightarrow$

$$\Rightarrow |f(x) - L| < \epsilon \quad \text{implied.}$$

$$\lim_{x \rightarrow 2} x^2 = 4$$

$$f(x) = x^2, \quad L = 4, \quad x_0 = 2.$$

show: $\forall \epsilon > 0 \quad \exists \delta > 0 \quad \text{s.t.} : \boxed{0 < |x - 2| < \delta} \quad \text{implies}$
 $|x^2 - 4| < \epsilon.$

$$\text{Solution: } |x^2 - 4| = |f(x) - L|$$

$\Downarrow$

$$|x^2 - 4| = |(x-2) \cdot (x+2)| =$$

$$= \underbrace{|x-2| \cdot |x+2|}$$

the factor $|x-2|$ is small (because by def $|x-2| < \delta$), The factor $|x+2|$ is not small $\Rightarrow$ But we can show that is bound.

bounding: $|x+2|$ - Let us say that $\delta \leq 1 \Rightarrow$

$$\Rightarrow \text{by the def we know: } \begin{cases} 0 < |x-2| < \delta \\ \delta \leq 1 \end{cases} \Rightarrow$$

$$\Rightarrow |x-2| \leq 1 \Rightarrow -1 \leq x-2 \leq 1$$

$$1 \leq x \leq 3 \Rightarrow$$

$$\Rightarrow |x+2| \leq 5$$

$$|x^2 - 4| = \underbrace{|x-2|}_{< \delta} \cdot \underbrace{|x+2|}_{\leq 5}$$

$$\begin{cases} |x^2 - 4| < \delta \cdot 5 \\ |x^2 - 4| < \epsilon \end{cases} \Rightarrow 5\delta = \epsilon \Rightarrow \delta = \frac{\epsilon}{5}.$$

$$\epsilon > 0$$

$$\text{Ans: } \delta = \min \left\{ 1, \frac{\epsilon}{5} \right\}^+$$

$$\text{Verification: } |x^2 - 4| = |x-2| |x+2| < \delta \cdot 5 \leq \frac{\epsilon}{5} \cdot 5 = \epsilon$$

$$\underline{\underline{|x^2 - 4| < \epsilon.}}$$

Problem 5.

$$\text{Def (Heine)} \quad \lim_{x \rightarrow x_0} f(x) = L \Leftrightarrow \forall \{x_n\}: x_n \in U(x_0), \lim_{n \rightarrow \infty} x_n = x_0 \Rightarrow \left[\lim_{n \rightarrow \infty} f(x_n) = L \right]$$

Negation rule: To negate a statement $\Rightarrow$ replace $\forall \leftrightarrow \exists$

$$\text{Solution: } \lim_{x \rightarrow 0} \sin\left(\frac{1}{x}\right) \rightarrow \text{DNE}$$

By the negation we need to prove:

$$\exists \{x_n\}: x_n \in U(0), \lim_{n \rightarrow \infty} x_n = 0, \text{ but } \lim_{n \rightarrow \infty} f(x_n) \neq L$$

$$\underbrace{x_n = \frac{1}{n}}_{=0} \quad f(x_n) = \sin\left(\frac{1}{x}\right)$$

$$\checkmark \lim_{x \rightarrow 0} x_n = 0, \quad x_n = \frac{1}{n}, \quad L = 0, \quad \forall \epsilon > 0$$

(by ϵ -def)

$$\lim_{x \rightarrow 0} f(x_n) = L, \quad \text{assume that } L \text{ exists}$$

$L = a \text{ value.}$

$$f(x_n) = \sin\left(\frac{1}{x}\right)$$

$x \rightarrow \infty$

$$\sin\left(\frac{1}{\infty}\right) = 0 \quad (\text{by def})$$

$$\sin\left(\frac{1}{x}\right) = 1$$

$$x_n = \frac{1}{2\pi n + \pi/2}$$

$$\lim_{n \rightarrow \infty} f(x_n) = 1$$

$$\sin\left(\frac{1}{x}\right) = -1$$

$$x_n = \frac{1}{2\pi n - \pi/2}$$

$$\lim_{n \rightarrow \infty} f(x_n) = -1.$$

$\Rightarrow \lim \rightarrow \text{DNE.}$

Problem 7.

$x \rightarrow x_0$ type of ind = $\left\{ \frac{0}{0} \right\}$.

$$a) \lim_{x \rightarrow 0} \frac{x^2 + 3x}{x} = \frac{0^2 + 3 \cdot 0}{0} = \left\{ \frac{0}{0} \right\} = \lim_{x \rightarrow 0} \frac{\cancel{x}(x+3)}{\cancel{x}} =$$

$$\lim_{x \rightarrow 0} (x+3) = 3.$$

$$b) \lim_{x \rightarrow 2} \frac{4x^2 - 12x + 8}{x^2 + 3x - 10} = \frac{4 \cdot 2^2 - 12 \cdot 2 + 8}{2^2 + 3 \cdot 2 - 10} = \left\{ \frac{0}{0} \right\} =$$

$$= \lim_{x \rightarrow 2} \frac{4(x-1)\cancel{(x-2)}}{\cancel{(x-2)}(x+5)} = \lim_{x \rightarrow 2} \frac{4x-4}{x+5} = \frac{4}{7}$$

$$4x^2 - 12x + 8 = (x-2)q(x) + r_1(x)$$

$$x^2 + 3x - 10 = (x-2)p(x) + r_2(x)$$

$$c) \lim_{x \rightarrow 1} \frac{x^3 + 4x^2 - 5}{x^4 - 1} = \left\{ \frac{0}{0} \right\} = \lim_{x \rightarrow 1} \frac{\cancel{(x-1)}(x^2 + 5x + 5)}{\cancel{(x-1)}(x+1)(x^2+1)} =$$

$$x^3 + 4x^2 - 5 = (x-1)q(x) + r_1(x)$$

$$x^4 - 1 = (x-1)q(x) + r_2(x)$$

$$= \frac{11}{4}$$

$$d) \lim_{x \rightarrow 7} \frac{7-x}{\sqrt{x+9} - 4} = \left\{ \frac{0}{0} \right\} = \lim_{x \rightarrow 7} \frac{\cancel{(7-x)}(\sqrt{x+9} + 4)}{\underbrace{x+9 - 16}_{x-7 = -\cancel{(7-x)}}} =$$

$$= - \lim_{x \rightarrow 7} (\sqrt{x+9} + 4) = -8.$$

$$f) \lim_{x \rightarrow +\infty} \frac{2 \cdot 5^x - 3}{9 \cdot 5^x + 4} = \lim_{n \rightarrow \infty} \left(\frac{a}{b^n} \right) = 0$$

$$= \lim_{x \rightarrow +\infty} \frac{2 - \frac{3}{5^x}}{9 + \frac{4}{5^x}} = \frac{2}{9}$$

$$\lim_{x \rightarrow -\infty} \frac{2 \cdot 5^x - 3}{9 \cdot 5^x + 4} = \frac{2 \cdot 5^{-\infty} - 3}{9 \cdot 5^{-\infty} + 4} = -\frac{3}{4}$$